

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

5 Formal Models of Language (pjb48)

An intrepid linguist encounters an alien while exploring a remote planet, she documents an alien language that has four sounds, $\Sigma = \{a, e, f, m\}$. The following are examples of the alien's speech and a translation:

$e\ amaf$	nice spaceship
$e\ faem$	nice spaceboots
$e\ e\ amaf$	very nice spaceship
$e\ e\ e\ faem$	extremely nice spaceboots

The linguist hypothesises that Expression 1 matches sound sequences in the alien language.

Expression 1: $e^*(a|e|f|m)(a|e|f|m)^*$

Regular
Expression

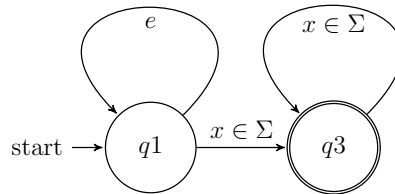
- (a) Describe the strings that are matched by Expression 1. [2 marks]

Answer: any number of e 's [1 mark] followed by $(a|e|f|m)(a|e|f|m)^*$ a non-zero combination of sounds [1 mark]

Finite State
Automata

- (b) Provide a Finite State Automaton that can generate the language defined by Expression 1. [2 marks]

Answer: [1 mark] for an FSA that matches the expression given [1 mark] for fully defined FSA (has a start, end states and labelled arcs).



As the linguist continues to explore, she encounters a grumpy alien and documents the following utterances:

$fama\ e\ amaf$	not nice spaceship (unpleasant spaceship)
$meaf\ e\ e\ faem$	very unpleasant spaceboots
$afaf\ e\ e\ e\ fafa$	extremely unpleasant gift

The linguist hypothesises that Expression 2 matches sound sequences in the language.

Expression 2: $w^{-1}e^*w$ where $w \in \Sigma^*$

Pumping lemma
for Regular
Languages

- (c) Can the linguist draw a Finite State Automaton to generate the language defined by Expression 2? Provide a proof for your answer. [5 marks]

Answer: No. Proof: A language may be generated by a Finite State Automaton if it is regular. The pumping lemma for regular languages is used to prove that a language is not regular. The pumping lemma property is:

— *Solution notes* —

All $w \in \mathcal{L}$ with $|w| \geq l$ can be expressed as a concatenation of three strings, $w = u_1vu_2$, where u_1, v and u_2 satisfy:

- $|v| \geq 1$ (i.e. $v \neq \epsilon$)
- $|u_1v| \leq l$
- for all $n \geq 0$, $u_1v^n u_2 \in \mathcal{L}$

[2 marks]

To prove that $\mathcal{L} = \{w^{-1}e^*w\}$ is not regular find some $s \in \mathcal{L}$ of length $\geq l$ so that no matter how s is split into three, $s = u_1vu_2$, with $|u_1v| \leq l$ and $|v| \geq 1$, there is some $n \geq 0$ for which $u_1v^n u_2$ is not in $\mathcal{L}$. [1 mark]

Consider $a^l e a^l \in \mathcal{L}$ [1 mark] u_1v must consist entirely of a . In the case where $n = 0$ I will no longer have the same number of a 's either side of the e which is not in $\mathcal{L}$ [1 marks]

Context free
grammar

- (d) Provide a grammar that can generate the language defined by Expression 2. [5 marks]

Answer: Providing a full CFG specification. [1 mark]

$$\begin{aligned} \mathcal{N} &= \{S, X, Y\} \\ \Sigma &= \{e, a, m, f\} \\ S &= S \\ \mathcal{P} &= \{S \rightarrow Y \mid X \quad [2 \text{ marks}] \\ &\quad Y \rightarrow e S e \mid a S a \mid m S m \mid f S f \quad [1 \text{ mark}] \\ &\quad X \rightarrow \epsilon \mid e \mid X e\} \quad [1 \text{ mark}] \end{aligned}$$

On the far side of the planet, the linguist encounters a new dialect and documents the following utterances:

<i>amaf e amaf</i>	unpleasant spaceship
<i>faem e e faem</i>	very unpleasant spaceboots
<i>mefaem e e e mefaem</i>	extremely unpleasant earthling

Characterising
strings

- (e) Provide a general expression to match such sound sequences. [1 mark]

Answer: we^*w where $w \in \Sigma^*$ [1 mark]

Pumping lemma
for CFL

- (f) Can you use a Context Free Grammar to generate the language defined by your expression? Provide a proof for your answer. [5 marks]

Answer: No. The pumping lemma for context free languages (CFLs) is used to show that a language is not context free. The pumping lemma property for CFLs is:

All $w \in \mathcal{L}$ with $|w| \geq k$ can be expressed as a concatenation of five strings, $w = u_1yu_2zu_3$, where u_1, y, u_2, z and u_3 satisfy:

- $|yz| \geq 1$
- $|yu_2z| \leq k$
- for all $n \geq 0$, $u_1y^n u_2 z^n u_3 \in \mathcal{L}$

[2 marks]

— *Solution notes* —

To use the pumping lemma to prove that a language $\mathcal{L}$ is not context free for each $k \geq 1$, find some $s \in \mathcal{L}$ of length $\geq k$ so that no matter how s is split into five, $s = u_1 y u_2 z u_3$, with $|y u_2 z| \leq k$ and $|y z| \geq 1$, there is some $n \geq 0$ for which $u_1 y^n u_2 z^n u_3$ is not in $\mathcal{L}$ [1 mark].

Consider $s = ab^k eab^k \in \mathcal{L}$ [1 mark]. Case 1: $y u_2 z$ spans or is inside one of the b^k 's then $n = 0$ yield s not in $\mathcal{L}$. Case 2: $y u_2 z = ea$ giving three scenarios all of which yield s which not in $\mathcal{L}$ when $n = 0$. [1 mark]
